

Malik Hassanaly

Machine Learning & Uncertainty Quantification for Physics-Based Simulation
Denver, CO — on-site locally, or fully remote

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in [linkedin.com/in/malik-hassanalay](https://www.linkedin.com/in/malik-hassanalay)

🐙 github.com/malihass

🌐 malihass.github.io

🔗 Google Scholar

SUMMARY

Computational scientist working where machine learning meets physics-based simulation. I build the **AI/ML layer that sits on top of existing simulators** — uncertainty quantification, simulation-based inference, inverse parameter identification, uncertainty propagation, surrogate modeling — and ship it as software other people use. **I work across an unusually wide set of domains** (Li-ion batteries, atmospheric flows, power-system security, bioreactors, semiconductor manufacturing, gas turbines and engines). Project lead on a \$550k program, mentor to six junior researchers, author or maintainer of open-source tools with 300+ combined GitHub stars. PhD in Aerospace Engineering; 40+ peer-reviewed publications.

TECHNICAL EXPERTISE

- **Inference & Uncertainty:** Bayesian inference, simulation-based inference, inverse problems, uncertainty propagation, sensitivity analysis, rare-event probability estimation, Bayesian neural networks, normalizing flows, multi-fidelity methods, design of experiments
- **Machine Learning:** Generative models (GANs, diffusion-based models), physics-informed neural networks (PINNs), neural operators (DeepONet), surrogate and reduced-order modeling, reinforcement learning, anomaly detection, adversarial ML and automated red-teaming
- **Simulation & HPC:** Computational fluid dynamics, large-eddy simulation, reacting and multiphase flow, finite-volume solvers, GPU/CPU exascale computing, OpenFOAM, AMReX, ParaView
- **Software:** Python, C++, Bash, Git; PyTorch, TensorFlow; open-source package authorship, documentation, testing and release
- **Application Domains:** Li-ion batteries and energy storage, wind and solar resource modeling, electric power systems and grid security, bioreactors and process engineering, semiconductor manufacturing and combustion engines

PROFESSIONAL EXPERIENCE

- **National Laboratory of the Rockies (NLR)** September 2022 – Present
Golden, CO
Staff Researcher — AI & Physics Modeling
 - **Program lead, adversarial ML (\$550k).** Founded and lead a research program on automated red-teaming of power-system controllers: reinforcement learning to discover false-data-injection attacks, continual learning and generative anomaly detection. Published at ICLR 2025, IEEE PowerTech 2025, IEEE Green Tech 2026 and IEEE PES GM 2024.
 - **Scalable inverse inference for Li-ion batteries.** Replaced per-cell optimization with amortized Bayesian inference via neural posterior estimation, cutting parameter-identification cost from hours to milliseconds per cell and returning full posteriors instead of point estimates. Released as open-source *BatFIT*.
 - **Physics-informed surrogates for battery calibration.** Lead author on a two-part study of PINN surrogates for single-particle and pseudo-2D Li-ion models — multi-fidelity training hierarchies and regularization for stiff regimes — *Journal of Energy Storage* 2024. Shipped as *PINNSTRIPES* (60+ stars).
 - **Sample-efficient uncertainty propagation.** Built Bayesian-neural-network closure models that propagate input uncertainty through expensive simulators, making UQ affordable inside a design cycle rather than as an afterthought (*Engineering Applications of AI* 2025; *mluq-prop*).
 - **Design optimization and cost-aware experimental design.** Bayesian calibration of bubble-size dynamics in CO₂ gas fermenters and surrogate-based optimization of sparging and mixing in U-loop bioreactors. Open-sourced as *BioReactorDesign* (40+ stars).
 - **Scalable physics-based models.** New method for chemical kinetics integration on heterogeneous CPU/GPU exascale hardware (*Combustion and Flame* 2024, shipped in *PelePhysics*).
 - **Mentorship.** Mentored 4 graduate interns and 2 postdoctoral researchers. Received the Outstanding Mentor Award and the NLR President's Award.
- **National Laboratory of the Rockies (NLR)** September 2019 – September 2022
Golden, CO
Postdoctoral Researcher — AI & Physics Modeling
 - **Generative super-resolution with calibrated uncertainty.** Developed adversarial sampling of unknown, high-dimensional conditional distributions to super-resolve atmospheric flow fields while preserving ensemble diversity (*Journal of Computational Physics* 2022). Underpins *sup3r* (130+ stars), used to generate synthetic high-resolution wind and solar datasets.
 - **Intelligent data reduction.** Created iterative normalizing-flow sampling that selects training data uniformly in phase space, shrinking datasets for data-driven surrogates without losing rare regimes (*Data-Centric Engineering* 2023).
 - **Inverse modeling of surface chemistry and reactor optimization.** Calibrated surface-chemistry kinetics for GaAs epitaxial growth and hydride cracking against reacting-flow simulations, then used the calibrated model to optimize HVPE deposition reactor design (*Journal of Applied Physics* 2021).
- **University of Michigan** January 2015 – August 2019
Ann Arbor, MI
Graduate Research Assistant — Doctoral Research
 - **Numerical solver with real-world adoption.** Wrote a minimally-dissipative low-Mach solver for variable-density reacting flows in OpenFOAM, cutting numerical dissipation and mass-conservation error; **now used by 15+ institutions** (*Computers & Fluids* 2018).

- **Rare and extreme event characterization.** Produced the first estimate of the attractor dimension of turbulent flames using Lyapunov exponents, and identified extinction/reignition as the dominant driver of perturbation amplification — the foundation of my later rare-event probability estimation work.
- **Earlier industry experience** 2012 – 2013
Maïa Eolis (now Engie Green) · RTE, French national grid operator France
 - **Physics Modeling Intern, Maïa Eolis.** CFD modeling of a novel wind-turbine blade spoiler; quantified the induced mechanical stress. **Software Development Intern, RTE:** built a generic UML-to-C++ gateway for a large-scale power-grid simulation code.

SELECTED OPEN-SOURCE SOFTWARE

Full list: github.com/malihass

- **sup3r** (130+ stars) — contributor: Generative adversarial super-resolution for synthetic high-resolution wind and solar data.
- **PelePhysics** (80+ stars) — contributor: Chemistry and transport for the Pele exascale reacting-flow suite (2024 R&D 100 Finalist); authored the symbolic QSS Jacobian generator.
- **PINNSTRIPES** (60+ stars) — lead author: Physics-informed neural network surrogates of Li-ion battery models for parameter identification.
- **BioReactorDesign (BiRD)** (40+ stars) — lead author: OpenFOAM toolbox for bioreactor simulation, Bayesian calibration and design optimization.
- **BatFIT** — lead author; **BATMODS-lite** — contributor, on PyPI: Neural posterior estimation for Li-ion parameter inference; SPM and P2D cell models with NMC/LFP/graphite properties for synthetic training-data generation.
- **Phase-space-sampling** — lead author: Normalizing-flow data downselection for training-set reduction.
- **diversity_SR**, **mluq-prop**, **GANISP** — lead author: GAN diversity measurement and enforcement; ML uncertainty propagation; GAN-assisted rare-event probability estimation.

SELECTED PUBLICATIONS

40+ peer-reviewed publications — full list on [Google Scholar](#)

1. **Neural posterior estimation for scalable and accurate inverse parameter inference in Li-ion batteries**, M. Hassanaly et al., *Journal of Energy Storage*, 2026.
2. **PINN surrogate of Li-ion battery models for parameter inference, Parts I & II**, M. Hassanaly et al., *Journal of Energy Storage*, 2024.
3. **Swift Hydra: self-reinforcing generative framework for anomaly detection with multiple Mamba models**, N. H. K. Do, T. Nguyen, M. Hassanaly et al., *ICLR*, 2025.
4. **A priori uncertainty quantification of reacting turbulence closure models using Bayesian neural networks**, G. Pash, M. Hassanaly et al., *Engineering Applications of Artificial Intelligence*, 2025.
5. **Adversarial sampling of unknown and high-dimensional conditional distributions**, M. Hassanaly et al., *Journal of Computational Physics*, 2022.
6. **Bayesian calibration of bubble size dynamics applied to CO₂ gas fermenters**, M. Hassanaly et al., *Chemical Engineering Research and Design*, 2025.
7. **Uniform-in-phase-space data selection with iterative normalizing flows**, M. Hassanaly et al., *Data-Centric Engineering*, 2023.
8. **Bi-fidelity modeling of uncertain and partially unknown systems using DeepONets**, S. De, M. Reynolds, M. Hassanaly et al., *Computational Mechanics*, 2023.
9. **Classification and computation of extreme events in turbulent combustion**, M. Hassanaly et al., *Progress in Energy and Combustion Science*, 2021.

EDUCATION

- **PhD, Aerospace Engineering** — University of Michigan, 2019. Dissertation: “[Extreme Events in Turbulent Combustion](#)”
- **MSE, Aerospace Engineering** — University of Texas at Austin, 2015. Thesis: “[Large-eddy simulations of boundary layer flashback](#)”
- **MSE, General Engineering** — École Centrale de Lille, France, 2015. Thesis project: design and testing of a new tidal turbine

HONORS, AWARDS & SERVICE

- **2026:** R&D 100 Winner (Pele exascale simulation suite)
- **2024:** President’s Award · Outstanding Mentor Award
- **2022:** Milton Van Dyke Video Award — “Simulation of an RCCI Engine Using the Pele Suite of Exascale Codes”
- **2019:** Richard and Eleanor Towner Prize for Distinguished Academic Achievement
- **Service:** Proposal reviewer (DOE ASCR, LDRD, HPC allocations). Journal reviewer in AI/ML, numerical methods, inverse methods, UQ, battery and combustion modeling. Member: SIAM, APS, AIChE, AIAA, The Combustion Institute.
- **Languages:** English (fluent) · French (native) · Spanish (intermediate)